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Review

Comparison of different sewage sludge pretreatment technologies for improving sludge solubilization and anaerobic digestion efficiency: A comprehensive review

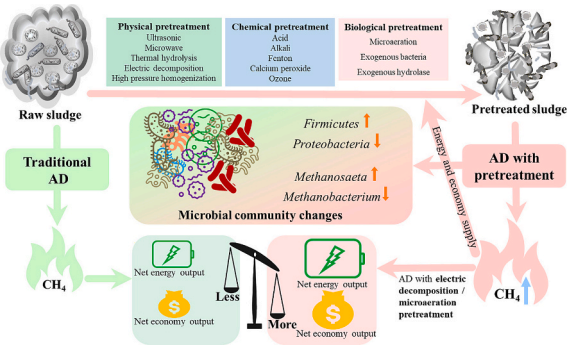
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HIGHLIGHTS

- Enhancing effects of mono pretreatments on sludge solubilization and AD efficiency are compared.
- Physical pretreatment excels in sludge solubilization, while biological pretreatment outperforms in enhancing AD efficiency.
- Most pretreatments elevate the relative abundance of *Firmicutes* and *Methanosaeta*.
- Electric decomposition and microaeration pretreatments stand out in energy and economic outcomes.
- A full energy and economic evaluation route for pretreatment-AD systems is proposed.

GRAPHICAL ABSTRACT



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ABSTRACT

Anaerobic digestion (AD) of sewage sludge reduces organic solids and produces methane, but the complex nature of sludge, especially the difficulty in solubilization, limits AD efficiency. Pretreatments, by destroying sludge structure and promoting disintegration and hydrolysis, are valuable strategies to enhance AD performance. There is a plethora of reviews on sludge pretreatments, however, quantitative comparisons from multiple perspectives across different pretreatments remain scarce. This review categorized various pretreatments into three groups: Physical (ultrasonic, microwave, thermal hydrolysis, electric decomposition, and high pressure homogenization), chemical (acid, alkali, Fenton, calcium peroxide, and ozone), and biological (microaeration, exogenous bacteria, and exogenous hydrolase) pretreatments. The optimal conditions of various pretreatments and their impacts on enhancing AD efficiency were summarized; the effects of different pretreatments on microbial community in the AD system were comprehensively compared. The quantitative comparison based on dissolution degree of COD (DD_{COD}) indicted that the sludge solubilization performance is in the order of physical, chemical, and biological pretreatments, although with each below 40 % DD_{COD}. Biological pretreatment, particularly microaeration and exogenous bacteria, excel in AD enhancement. Pretreatments alter microbial ecology, favoring *Firmicutes* and *Methanosaeta* (acetotrophic methanogens) over *Proteobacteria* and *Methanobacterium* (hydrogenotrophic

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methanogens). Most pretreatments have unfavorable energy and economic outcomes, with electric decomposition and microaeration being exceptions. On the basis of the overview of the above pretreatments, a full energy and economy assessment for sewage sludge treatment was suggested. Finally, challenges associated with sludge pretreatments and AD were analyzed, and future research directions were proposed. This review may broaden comprehension of sludge pretreatments and AD, and provide an objective basis for the selection of sludge pretreatment technologies.

1. Introduction

Sewage sludge, the byproduct of sewage treatment, enriches organic matters, solid particles, metal salts, and other hazardous compounds in sewage (Jiang et al., 2021). There are many beneficial substances in sewage sludge that can be recycled prior to incineration or landfill disposal. Notably, the biological organic matters, encompassing proteins, polysaccharides, and lipids et al., hold the potential for biomass energy recycling (Li et al., 2022). However, these organic components are either directly present in microbial cells or encapsulated in the stable structure of sewage sludge (Xu et al., 2020a), so how to effectively harness the biological organic matters in sewage sludge has posed a formidable global challenge. Anaerobic digestion (AD) is a highly valuable technique for sludge reduction and resource recovery, as it reduces the solids content in sewage sludge and facilitates the recovery of methane as energy (Xu et al., 2020a; Stasinakis, 2012).

Biogas or methane yield and the removal of volatile solid (VS) and total chemical oxygen demand (TCOD) in AD are crucial indicators to evaluate AD efficiency (Fitamo et al., 2017; Khanh Nguyen et al., 2021; Zhang et al., 2021). Traditional AD efficiency of sewage sludge is limited, with insufficient methane production and inadequate VS and TCOD removal, for the two reasons listed below. Firstly, the intricate composition and robust structure of sewage sludge impede AD efficiency. The stable structure of sewage sludge acts as a barrier to the dissolution of organic matters (Khanh Nguyen et al., 2021). Given the challenge posed by dissolving intracellular organic matters and extracellular polymeric substances (EPS), hydrolysis is the rate-limiting phase in the AD of sewage sludge. Furthermore, the presence of heavy metals, inorganic particles, emerging micropollutants, and refractory organic substances diminishes the activity of functional microorganisms, thereby impairing AD performance (Stasinakis, 2012; Luo et al., 2020; Wu et al., 2022). Secondly, the inherent technical limitations of AD, such as long operation cycle and rigorous operating conditions, contribute to its suboptimal performance. The sludge retention time (SRT) of AD is typically 20–30 days due to the long generation cycle of methanogens, challenging to further shorten SRT in traditional AD (Liu et al., 2020a). Furthermore, because of the sensitivity of methanogens to the environment, the fluctuations of temperature, redox potential, and pH in AD systems may induce a decline in their activity, subsequently reducing AD efficiency (Lin et al., 2017; Yan et al., 2018).

Various strategies have been explored to improve the AD efficiency of sewage sludge. Pretreatments (Khanh Nguyen et al., 2021), co-digestion (Elalami et al., 2019), two-phase AD (Nabaterega et al., 2021), AD assisted with microbial electrolysis cell (Wang et al., 2022a), addition of conductive substances (Kumar et al., 2021; Wei et al., 2018a; Wang et al., 2024), and optimizing process conditions (temperature, load, SRT, mixing) (Liu et al., 2020a; Kariyama et al., 2018) are common options. Among them, pretreatment technology can extensively improve sludge solubilization, and augment the content of biodegradable organic matters in the liquid phase. There are three categories of pretreatment: physical, chemical, and biological pretreatments. Physical pretreatment, such as ultrasound, thermal hydrolysis, and electric decomposition et al., can promote sludge decomposition and microbial cell destruction through various physical effects, while stimulating the release of organics from sewage sludge (Tyagi and Lo, 2011). Chemical pretreatment can inactivate microbial cells and result in sludge degradation by adding chemicals such as strong acid, alkali, and oxidants. Biological pretreatment relies on the biological catalysis of hydrolases to hasten the decomposition of organic matters within sludge; hydrolases can be sourced from endogenous facultative bacteria that secretes hydrolases under microaerobic condition, exogenous bacteria that efficiently secretes specific hydrolases, and the direct addition of hydrolases.

There already have been some reviews on sludge pretreatments. Some reviews summarized the principles, conditions, and efficiency of individual pretreatments, such as ultrasonic (Pilli et al., 2011; Bhat and Gogate, 2021), high pressure homogenization (Nabi et al., 2020), thermal hydrolysis (Barber, 2016; Ngo et al., 2021), the advanced oxidation of hydrogen peroxide and Fenton (Guan et al., 2018; Pilli et al., 2015), and microaeration (Fu et al., 2023). Other reviews categorized and examined various pretreatments, providing the pretreatment status and identifying future research avenues. (Khanh Nguyen et al., 2021; Carrere et al., 2010; Zhen et al., 2017). In addition, the feasibility of pretreatments in the energy balance was also examined (Cano et al., 2015). However, comparative analysis of inherent characteristics of different pretreatments and their impacts on sludge solubilization and AD efficiency enhancement need to be strengthened in order to identify acceptable pretreatments, as well as their optimal conditions.

This review classifies numerous pretreatments into physical, chemical, and biological categories, and summarizes the mechanisms, conditions, and performance of each pretreatment. Innovatively, different pretreatments are quantitatively compared for pretreatment performance, AD efficiency enhancement, microbial community change, and energy and economy assessments. To compare different mono pretreatments more intuitively, combined pretreatments are not considered. We expect that this review will provide valuable guidance in the selection and optimization of sludge pretreatments.

2. Physical pretreatment

2.1. Ultrasonic pretreatment

Ultrasonic pretreatment induces periodic compression and sparseness of sewage sludge, causing cavitation effect characterized by immediate high temperature, high pressure, strong fluid shear, and the generation of active free radicals like H· and ·OH (Zhen et al., 2017; Zheng et al., 2023). The cavitation phenomenon damages and disintegrates microbial cells and EPS of sewage sludge, raising the soluble chemical oxygen demand (SCOD) by dozens of times and considerably improving methane yield in AD (Apul and Sanin, 2010; Li et al., 2018). In addition to enhancing the dissolution of proteins and polysaccharides, ultrasonic pretreatment also promotes the release of extracellular hydrolases. (Zheng et al., 2023; Yu et al., 2008). Moreover, the activity and quantity of enzymes such as protease, α-amylase, and α-glucosidase exhibit an increase proportional to the duration and energy density of ultrasonic pretreatment (Yu et al., 2008). The released enzymes and their elevated activities after ultrasonic pretreatment also contribute to some extent to the hydrolysis of dissolved organic matter and organic matter in the sludge surface layer. However, Yu et al. (2008) found the enzymes were released more in the slime than in the supernatant after ultrasonic pretreatment, and the activities of enzymes (protease, a-amylase, alkaline-phosphatase, acid-phosphatase) decreased more and faster during AD stage than that in the unpretreated sludge. Therefore, the main contributor to the increase in methane yield during AD stage is not the elevated enzymatic activity, but the increased SCOD.

As shown in Table 1, ultrasonic frequency and duration emerge as the most crucial factors. (Bhat and Gogate, 2021; Le et al., 2015). The cavitation effect, characterized by the critical size of cavitation bubbles, is influenced by the ultrasonic frequency, typically ranging from 20 to 25 kHz. Although the sludge solubilization increases with extended ultrasonic duration, it should not surpass 15 min, as the solubilization improves less beyond this threshold (Apul and Sanin, 2010; Li et al., 2018; Martin et al., 2015; Şenol, 2021; Sahinkaya, 2015). For instance, while the SCOD increase was around 45.6 % at 15 min, it was 67.9 % at 240 min (Şenol, 2021). Moreover, the prolonged duration will result in higher energy consumption. Notably, ultrasonic pretreatment leads to an increase in refractory organics in the liquid phase, such as humic substances and complex high molecular weight proteins, which may inhibit the activity of coenzyme F420 (a key enzyme in methanogenesis) (Zheng et al., 2023; Lu et al., 2018a), and these substances increase in quantity with longer ultrasonic duration. In a semi-continuous AD system, the performance of ultrasonic pretreatment also hinges on the specific energy input and the organic load, and the biogas yield increased with the upgrading of organic load (1–3 kg VS/m^3·d) at high specific energy (35,000 kJ/kg TS), with the optimal yield of 548 mL/g $VS_{removed}$ at lower SRT of 6.7 d (Lizama et al., 2018). A strong positive correlation between specific energy input and the dissolution degree of sewage sludge was found (Lu et al., 2018a). In general, ultrasonic pretreatment has a notable performance in sludge disintegration and AD enhancement, and it would be a promising strategy if the energy consumption could be further reduced.

2.2. Microwave pretreatment

In microwave pretreatment, both thermal and non-thermal effects (including electrical, magnetic, and chemical effects) are employed to disrupt EPS and microbial cells within sludge (Zhen et al., 2017). Table 1 summarizes the effects of power, duration, and temperature on the performance of microwave pretreatment. Gil et al. (2018) observed 34.6 % enhancement in sludge biodegradability and 20 % increase in methane yield within a semi-continuous AD systems utilizing microwave pretreatment with 700 W, 63 s, and 80 °C. Additionally, microwave power and duration determine the specific energy input. A review conducted by Bozkurt and Apul (2020) outlined that SCOD increase (2 % to 170 %) exhibited positive linear relationship with the specific energy input of below 300 kJ/g VS, but it sharply increased to 1000 %–2000 % when the specific energy input rose from 300 to 8000 kJ/g VS. Moreover, due to the significance of the thermal effect, the temperature of microwave pretreatment has a greater impact on the dissolution of organic matters compared to the duration (Kor-Bicakci et al., 2019; Yu et al., 2010). Nevertheless, increasing the microwave temperature does not always significantly improve methane yield in AD, potentially due to the generation of more refractory compounds through thermal effects (Toreci et al., 2009). It has been reported that merely microwave-treated sewage sludge was challenging to energy balance in the AD systems, and it even resulted in larger negative net energy output at high temperatures (Kor-Bicakci et al., 2019; Alqaralleh et al., 2019). Enhancing TS content of sewage sludge (up to 125 g/L) may potentially reduce the energy loss (Zhang et al., 2016; Liu et al., 2020b). Given the AD efficiency and energy consumption, it is recommended that the suitable conditions for microwave pretreatment are: A specific energy input of 20–30 kJ/g TS, a temperature not exceeding 100 °C, and a duration of less than 10 min (Gil et al., 2018; Bozkurt and Apul, 2020; Kor-Bicakci et al., 2019; Park et al., 2004; Eskicioglu et al., 2007).

2.3. Thermal hydrolysis pretreatment

Thermal hydrolysis can be used for the pretreatment of sewage sludge with a high level of TS (13–21 %, *w*/w), which helps to reduce unnecessary heat loss (Lu et al., 2018b; Zhang et al., 2017; Feng et al., 2015). Significantly, the pretreatment of high-solid sludge through thermal hydrolysis not only addresses the challenge of low mass transfer rates but also enhances the volumetric utilization rate of AD devices (Choi et al., 2018).

Temperature and duration represent the primary factors influencing sludge properties during thermal hydrolysis. An overview of thermal hydrolysis performance is provided in Table 1. Typically, high-temperature thermal hydrolysis should not exceed 180 °C for less than 1 h, while low-temperature thermal hydrolysis should not exceed 90 °C for less than 24 h. Interestingly, the temperature exerts a stronger influence on AD efficiency enhancement in high-temperature thermal hydrolysis compared to the duration, while the duration plays a more critical role in low-temperature thermal hydrolysis (Zhang et al., 2017; Abelleira-Pereira et al., 2015). Notably, low-temperature thermal hydrolysis even exhibits positive energy balance compared with microwave or ultrasonic pretreatments (Balasundaram et al., 2022). Although thermal hydrolysis substantially elevates the SCOD of sewage sludge, the increase in methane production during AD is not high enough to match the significant augment of SCOD. For example, after low-temperature thermal hydrolysis for 24 h at 60 °C, 70 °C, 80 °C, and 90 °C, SCOD/TCOD increased from 4.5 % (raw sludge) to 29.6 %, 30.3 %, 34.8 %, and 41.1 %, respectively, while biogas production increased only by 11.3 %, 9.3 %, 12.3 %, and 8.8 % (Xue et al., 2015). This incongruity is attributed to the presence of Maillard products and the greater release of recalcitrant or inhibitory compounds such as humic substances, steroid-like matters, aromatics, benzenoids, flavonoids, pyridines, and their derivatives (Lu et al., 2018b; Wang et al., 2021; Huang et al., 2021a). These substances diminish the biodegradability of pretreated sludge, and even inhibit the activity of key enzymes (methyl-CoM reductase, acetate thiokinase, and coenzyme F420) in methanogenesis. Thus, future research efforts could be directed toward either reducing the production of these substances during the thermal hydrolysis pretreatment, or enhancing their degradation and utilization.

2.4. Electric decomposition pretreatment

Electric decomposition, characterized by its simplicity of operation, has garnered substantial research attention, which serves to augment the permeability of microbial cell membrane and even rupture the cell wall (Zeng et al., 2019; Ye et al., 2016). The electric decomposition process contributes to the release of lysozymes and hydrolases, ultimately enhancing the SCOD (Yu et al., 2018). Parameters such as electrode distance, voltage, current density, and electrolysis time (Table 1), collectively dictate the performance of electric decomposition (Paul Choudhury and Kalamdhad, 2021; Huang et al., 2021b; Xi et al., 2023). Yang et al. (2014) observed a 33 % increase in SCOD for concentrated waste activated sludge following electric decomposition pretreatment under the conditions of 4 mm electrode gap, 60 V, 9.45 mA cm^{-2}, and 30 min, accompanied by a 44 % rise in methane production in AD. Of these parameters, voltage emerges as the pivotal parameter (Zeng et al., 2019). In a study by Ye et al. (2016), the SCOD exhibited a linear increase from 141 to 2356 mg/L as voltage escalated from 0 to 20 V. Recent research indicated that voltage triggers the polarization of organic functional groups in sewage sludge, such as amide groups (Wang et al., 2022b). This polarization fosters extracellular interspecies electron transfer between functional microorganisms *Methanothrix* and *Firmicutes*, accelerating the methanogenetic rate. Moreover, electric decomposition of sewage sludge can be directly combined with AD systems in situ to construct a microbial electrolysis cells-AD systems to promote direct interspecific electron transfer of electroactive bacteria and electrotrophic methanogen, enhancing methanogenic efficiency (Zakaria and Ranjan Dhar, 2022; Lee et al., 2017).

In comparison to other physical pretreatments, electric decomposition minimizes the introduction of inhibitors for methanogenesis, and has potential application prospect with its notable advantages of positive net energy output and low operational cost. However, the electrodes used in this pretreatment mostly are platinum, titanium, ruthenium

Table 1
Enhancement of sludge solubilization and AD efficiency by various mono physical pretreatments.

Pretreatment	Sludge type	Raw sludge property	Pretreatment conditions	Pretreatment performance	Enhancement of AD efficiency	References
Ultrasonic	WAS	15.9 g TSS/L, 14.6 g VSS/L; 20.94 g TCOD/L, 68 mg SCOD/L	20 kHz, 25 °C, 10 min, 3 KW/L	TSS and VSS decreased by 11.8 % and 6.9 %		(Yu et al., 2008)
Ultrasonic	WAS	3.3–4.0 % TS	9 kHz, 20–25 °C, 40 min	Organic matter destruction efficiency increased by 48.1 %	Methane production increased by 69 %	(Wang et al., 1999)
Ultrasonic	WAS	3.76 g VS/L	24 kHz, 15 min, 0.51 W/mL	SCOD increased by 49 times; VS decreased by 8 %	VS removal increased by 28 %; Methane yield increased by 95.4 % (239 → 467 mL/g VS $_{removed}$); Methane content increased by 25 %	(Apul and Sanin, 2010)
Ultrasonic	Sewage sludge	132 g TS/kg, 88 g VS/kg	25 °C, 45 min, 775 J/g VS	VFA, TOC, TN increased by 39.8 %, 81.5 % and 50.0 %	VS removal was up to 81 %; Methane yield increased by 95.5 % (88 → 172 mL/g VS)	(Martin et al., 2015)
Ultrasonic	WAS	25.85 g TCOD/L, 338.0 mg SCOD/L; 25.2 g TS/L, 21.7 g VS/L	20 kHz, 80 min, 0.5 W/mL	DD_{COD} = 30.6 %	TCOD removal increased by 37.1 %; TS and VS removal increased by 24.0 % and 26.8 %; Methane yield was 36.2 mL/g VS; Methane content was more than 53.8 %	(Li et al., 2018)
Ultrasonic	Mixture of primary sludge and WAS	38.5 g TS/L, 29.2 g VS/L; 2045 mg SCOD/L	20 kHz, 30 min, 0.5 W/mL	SCOD increased by 60.5 %	Methane yield increased by 52.7 % (170 → 266 mL/g VS)	(Şenol, 2021)
Ultrasonic	WAS	38.40 g TCOD/L, 1041 mg SCOD/L; 2.89 % TS, 1.9 % VS	25,000 KJ/Kg TS		VS removal ranged from 17.27 % to 23.7 %; Biogas yield increased by 143.3 % (240 → 584 mL/g VS $_{removed}$); Methane content increased by 12.2 %	(Lizama et al., 2018)
Ultrasonic	WAS	28.49 g TS/L, 13.85 g VS/L; 18.96 g TCOD/L, 221.3 mg SCOD/L	20–25KHz, 1 W/mL, 15 min	DD_{COD} = 2.48 %	Methane yield increased by 15.3 % (108.72 → 125.39 mL/g VS)	(Zheng et al., 2023)
Ultrasonic	WAS	33.81 g TS/L, 24.87 g VS/L	24 kHz, 0.73 W/mL for 10 min	SCOD increased by 340 %	Methane yield increased by 32 % (104.8 → 138.5 mL/g VS)	(Çelebi et al., 2021)
Microwave	WAS	7.17 % TS, 5.39%VS; 74.07 g TCOD/L, 2307 mg SCOD/L	1000 W, 2 min, 0.83 KJ/mL		Methane yield increased by 15.4 % (195 → 225 mL/g VS); Methane content stabilized at 62.94 %	(Solyom et al., 2011)
Microwave	WAS	5.4 % TS, VS/TS = 69.8 %	96 °C, 1250 W, 2450 MHz	SCOD/TCOD increased by 360 %	Biogas yield increased by 16.7 %; Methane content ranged from 65 % to 81 %	(Eskicioglu et al., 2007)
Microwave	WAS	30.07 g TS/L, VS = 19.21 g VS/L	2450 MHz, 700 W, 15 min	DD_{COD} = 20.4 %	TCOD removal increased by 81.5 %; VS removal increased by 13.0 %; Methane yield increased by 30.2 % (242 → 315 mL/g VS); Methane content ranged from 72 to 76 %	(Park et al., 2004)
Microwave	Sewage sludge	27.46 g TCOD/L, 6.57 g SCOD/L; 21.94 g TS/L, 17.57 g VS/L	700 W, 63 s, 20 KJ/g TS	DD_{COD} = 3.4 %; Biodegradability increased by 34.6 %	Methane yield increased by 20 % (155 → 186 mL/g VS)	(Gil et al., 2018)
Microwave	Mixture of fermented primary sludge and WAS	3.94 % TS, 3.38 % VS; 15.51 g TCOD/L, 2107 mg SCOD/L	2450 MHz, 30 min, 160 °C, 28.12 KJ/g TS	DD_{COD} = 2.6 %	VS removal increased by 7.7 % and 19.1 %; Biogas yield increased by 18.6 % (489 → 580 mL/g VS) and 10.9 % (532 → 590 mL/g VS) at thermophilic and mesophilic AD	(Kor-Bicakci et al., 2019)
Microwave	WAS	3.33 % TS, 1.58 % VS; 13.92 g TCOD/L	2450 MHz, 175 °C	DD_{COD} = 20.9 %	Biogas yield increased by 3.4 % (359.2 → 394.8 mL/g VS); Methane content increased by 3.9 %	(Toreci et al., 2009)
Thermal hydrolysis	Dewatered sludge	16.7 % TS, 70.5 % VS/TS; 166.0 g TCOD/L, 7.55 g SCOD/L	One: 80 °C and 24 h; Two: 180 °C and 3 h	DD_{COD} = 33.8 %, 51.2 %	Biogas yield increased by 8.3 % (960 → 1040 mL/g VS $_{removed}$), 11.5 % (960 → 1070 mL/g VS $_{removed}$); Methane content ranged from 65 to 70 %	(Xue et al., 2015)
Thermal hydrolysis	Dewatered sludge	14.2 % TS, VS/TS = 57.4 %; 715 mg TCOD/g TS	One: 90 °C and 60 h; Two: 180 °C and 60 min	DD_{COD} = 49.7 %		(Zhang et al., 2017)
Thermal hydrolysis	Dewatered sludge	101.97 g TS/L, 71.05 g VS/L; 93.56 g TCOD/L, 2.11 g SCOD/L	165 °C, 50 min	DD_{COD} = 46.3 %	VS removal increased by 131.9 %; Methane yield increased by 157.1 % (70 → 180 mL/g COD/d); Methane content increased by 12.5 %	(Han et al., 2017)
Thermal hydrolysis	Digested sludge	4.6 % TS, 51.4 % VS/TS	180 °C, 30 min	The solubilization of the digestate increased by 10.4 times	Methane yield increased by 57.3 times (3 → 175.2 mL/g COD)	(Cai et al., 2021)
Thermal hydrolysis	Dewatered sludge	14.3 %TS; 169.0 g TCOD/L, 1.7 g SCOD/L	150 °C, 60 min	DD_{COD} = 27.3 %	TCOD removal increased by 52.2 %; Methane yield increased by 32.8 % (174 → 231 mL/g COD)	(Choi et al., 2018)

(*continued on next page*)

Table 1 (*continued*)

Pretreatment	Sludge type	Raw sludge property	Pretreatment conditions	Pretreatment performance	Enhancement of AD efficiency	References
Thermal hydrolysis	WAS	30.79 g TCOD/L, SCOD/TCOD = 2 %	200 °C, 30 min	DD_{COD} = 38.7 %	TCOD removal increased by 159.6 %; Biogas yield increased by 88.7 % (138.4 → 261.1 mL/g COD); Methane content increased by 8.1 %	(Yang et al., 2010)
Thermal hydrolysis	Mixture of primary sludge and WAS	32.0 g TSS /L, 25.2 g VSS/L; 41.99 g TCOD/L, 2.67 g SCOD/L	170 °C, 30 min	TSS decreased by 34.1 %, VSS decreased by 40.9 %; DD_{COD} = 27.8 %		(Hosseini Koupaie et al., 2021)
Electric decomposition	Petroleum sludge	11.44 % TS, VS/TS = 86.11 %; 1120 mg SCOD/L	60 V, 11.6 cm, 83.5 min	SCOD increased by 2.3 times	VS removal increased by 703.2 %; Methane yield increased by 66.7 % (3965 → 6610 mL/g VS)	(Paul Choudhury and Kalamdhad, 2021)
Electric decomposition	WAS	7.68 g TSS/L, 5.92 g VSS/L; 7.17 g TCOD/L, 210 mg SCOD/L	60 V, 4 mm, 9.45 mA/cm^2, 60 min	DD_{COD} = 20.6 %	Methane yield increased by 44.3 % (343 → 495 mL/g VS)	(Yang et al., 2014)
Electric decomposition	WAS	5.18 g TSS/L, 4.23 g VSS/L	15 V, 60 min	VSS removal increased by 20.3 %; SCOD increased by 11.9 times		(Zeng et al., 2019)
Electric decomposition	WAS	27.54–35.50 g TCOD/L, 141–502 mg SCOD/L; 26.75–31.56 g TS/g, 18.98–23.03 g VS/g	20 V, 5 cm, 30 min	TS removal = 1.51 %, VS removal = 2.89 %; SCOD increased by 7.0 times	TS removal and VS removal increased by 3.9 %, 14.3 %; TCOD removal increased by 15.0 %; Methane yield increased by 20.47 % (171 → 206 mL/g VS); Methane content increased by 9.0 %	(Ye et al., 2016)
Electric decomposition	WAS	7.4 g TSS/L, 6.4 g VSS/L; 12.66 g TCOD/L, 1331 mg SCOD/L;	20 V, 4 cm, 40 min	DD_{COD} = 8.2 %; TS removal and VS removal increased by 24.4 %, 32.1 %		(Zhen et al., 2014)
High pressure homogenization	WAS	8.425 g TCOD/L, 192 mg SCOD/L; 6.540 g TSS/L 4.675 g VSS/L	60 MPa, three cycles	DD_{COD} = 37.87 %; VSS decreased by about 59 %		(Fang et al., 2015)
High pressure homogenization	WAS	12.196 g TS/L, 7.103 g VS/L; 7.153 g TCOD/L, 120 mg SCOD/L	60 MPa	DD_{COD} = 31 %	VS removal increased by 30 %; Methane yield increased by 94.8 % (77 → 150 mL/g VS); Methane content increased by 42.2 %	(Nabi et al., 2019)
High pressure homogenization	Sewage sludge from MBR	23.67 g TS/L, 15.47 g VS/L; 22.37 g TCOD/L, 352.3 mg SCOD/L	50 MPa, two cycles	DD_{COD} = 7.7 %	VS removal increased by 138.4 %; TCOD removal increased by 274.2 %; Biogas production increased by 115.4 % after 7 days AD (1546 → 3330 mL); Methane content increased by 36.2 %	(Zhang et al., 2012a)
High pressure homogenization	Sewage sludge	14.716 g TS/L; 8.534 g VS/L; 13.342 g TCOD/L, 175 mg SOCD/L	60 MPa	DD_{COD} = 24 %;	Biogas yield increased by 11.6 % (215 → 240 mL/g COD/d); Methane content increased by 9.6 %; TCOD removal increased by 34.9 %; VS removal increased by 22.9 %	(Nabi et al., 2021)

oxide, yttrium oxide, or homemade composite conductive materials, which are expensive materials, and the initial cost needs to be considered in the scale-up application. In addition, the conditions of electric decomposition as well as the construction of the equipment vary considerably in different literatures (Yang et al., 2014; Ye et al., 2016), so it is difficult to determine the process parameters, especially the voltage, current density, and electrode distance. Therefore, it is worth studying how to promote this technology to engineering applications on the premise of pretreatment effectiveness and operational safety.

2.5. High pressure homogenization pretreatment

Recently, high pressure homogenization has emerged as a prominent avenue for sludge pretreatment, which harnesses the effects of high pressure, turbulence, cavitation, and robust shear within high pressure homogenizers, inducing significant alterations in sludge properties (Nabi et al., 2020). Specifically, the application of high pressure homogenization leads to a decrease in the average particle size, zeta potential, surface tension, and viscosity of sewage sludge, while concurrently increasing specific surface area, SCOD, and VFAs (Fang et al., 2015; Nabi et al., 2019). These transformations render the pretreated sludge more favorable to the proliferation and metabolism of anaerobic microorganisms. Generally, the performance of high pressure homogenization depends upon pressure, homogenization cycles, and solid content of sewage sludge (Table 1) (Zhang et al., 2012a; Zhang et al., 2012b). Pressure typically remains below 60 MPa, with cycles generally restricted to a maximum of three (Fang et al., 2015; Zhang et al., 2012a). In the study of Nabi et al. (2019), the escalation of pressure from 20 to 60 MPa yielded substantial improvements in SCOD, methane yield, and methane content, ascending from 1053 to 2342 mg/L, 77 to 150 mL/g VS, and 47 % to 64 %, respectively. To obtain optimal results, the solid content of sewage sludge introduced into high pressure homogenizers is recommended to remain below 25 g TS/L (Nabi et al., 2020; Zhang et al., 2012a; Zhang et al., 2012b; Nabi et al., 2021). The significant effect of high pressure homogenization pretreatment on the enhancement of AD of sewage sludge is conducive to the positive net energy output (Wahidunnabi and Eskicioglu, 2014), but its practical application may be limited by the technical difficulties of large-scale equipments with high pressure.

3. Chemical pretreatment

3.1. Acid and alkali pretreatments

Studies on acid and alkali pretreatments of sewage sludge to enhance AD efficiency are summarized in Table 2. Under acidic conditions, the

dissolution of protein and polysaccharide in EPS and microbial cells is enhanced as pH decreases (Devlin et al., 2011). Devlin et al. (2011) observed about 3 times increase in SCOD after acid (pH = 2) pretreatment of WAS, with 14.3 % increase in methane yield during AD. Remarkably, isoelectric point (pH = 2.6) pretreatment of WAS contributed to about 120 % enhancement in methane yield when compared to unpretreated WAS (Xu et al., 2022; Xu et al., 2021a). Recently extensive researches have been conducted on free nitrous acid (FNA) pretreatment of sewage sludge, and it was found that the proportion of biocidal cells rose as the FNA content increased, resulting in an increase in SCOD and AD efficiency (Ma et al., 2015; Zahedi et al., 2016; Wang et al., 2013; Meng et al., 2020). Specifically, within a low FNA content range (below 2.5 mg N/L), methane yield in AD was enhanced by 25–27 %, and in a higher FNA content range (4.9–6.5 mg N/L), the increase of methane yield could reach up to 37 % (Zahedi et al., 2016; Wang et al., 2013; Meng et al., 2020). Importantly, FNA can be derived from AD effluent, which can reduce the reagent cost and the energy consumption for sludge pretreatment (Wang et al., 2013; Wei et al., 2018b).

Under alkaline conditions, the optimal sludge solubilization occurs at pH of 11–12.5 (Xiao et al., 2015). Xu et al. (2020b) observed significant enhancements in soluble protein, polysaccharide, and SCOD by 9.82, 12.16, and 16.11 times, respectively, after alkali pretreatment at pH of 12, and this treatment resulted in 50 % increase in methane production during AD compared to the unpretreated group. To enhance economic viability, low pH alkali pretreatment has also garnered attention. Elalami et al. (2020) achieved 39 % enhancement in methane yield alongside 39 % VS removal in AD after alkali pretreatment at pH of 9. However, it is important to note that SCOD tends to increase with rising pH levels, yet the biodegradability of sewage sludge diminishes beyond pH of 9 (Wonglertarak and Wichitsathian, 2014). Consequently, alkali pretreatment with higher pH values may potentially hinder methanogenesis during AD (Shao et al., 2012). In addition to pH, the dose of alkaline reagents is another pivotal parameter (Chang et al., 2011). Among various options, NaOH is often preferred for alkali pretreatment due to its superior performance and lower chemical cost compared to $Ca(OH)_2$, $Mg(OH)_2$, and KOH (Toutian et al., 2021). The SCOD exhibits significant increment with the increase of NaOH dose ranging from 0 to 40 mg/g TS, while a limited increase is observed beyond 40 mg/g TS (Maryam et al., 2021; Zhang et al., 2015a). Furthermore, a downward trend of methane yield likely occurs at higher NaOH dose due to Na^+ inhibition for methanogens (Zhang et al., 2015a). The Na^+ inhibition mainly leads to an imbalance in cellular osmotic pressure, resulting in cell lysis and/or inactivation, as evidenced by the fact that higher concentration of Na^+ reduces the abundance and diversity of microorganisms, especially archaea (Wang et al., 2017). In addition, Na^+ inhibition also causes a decrease in microbial enzyme activities, such as coenzyme F420 and dehydrogenase (Zhang et al., 2020).

3.2. Fenton and calcium peroxide pretreatments

H_2O_2 is extensively utilized in sludge pretreatment via advanced oxidation processes, which often combines with ultrasonic, microwave, electricity, and Fe^{2+} catalyst to enhance oxidation capacity (Liu et al., 2017; Zhang et al., 2022). As summarized by Guan et al. (2018), H_2O_2 based advanced oxidation processes pretreatments significantly enhanced methane yield and VS removal in AD, reaching 147–323 mL/g VS and 33–42 %, respectively. Notably, Fenton agent, comprising H_2O_2 and Fe^{2+} catalyst (Table 2), can produce ample •OH under acidic conditions, boasting a higher redox potential than sole H_2O_2 or ozone (Pilli et al., 2015). Influencing parameters on Fenton pretreatment of sewage sludge have been explored by Pilli et al. (2015), covering H_2O_2 and iron content (0.01 mL H_2O_2/g SS, and $[H_2O_2]/[Fe^{2+}]$ ratio of 150), pH (3–6), temperature (20–40 °C), and reaction time (20 min). Methane yield in mesophilic AD was found to increase by 15 % (Pilli et al., 2016) and 26.9 % (Şahinkaya et al., 2015) after Fenton pretreatment. Thermophilic AD and two-stage AD were also improved with Fenton pretreatment, achieving 40 % and 30 % enhancement in methane yield, respectively (Erden and Filibeli, 2010). Remarkably, Fenton pretreatment of sewage sludge exhibited impressive energy balance performance, yielding 213 % improvement of net energy output over control (Pilli et al., 2016). Recent research also indicated that the Fenton-AD systems outperformed AD alone in energy balance ratio (energy output/input) by 42–50 % (Zhang et al., 2022).

Calcium peroxide (CaO_2) has greater stability and easy to be transported compared to H_2O_2, which provides both alkaline and oxidative conditions in sludge pretreatment (Luo et al., 2022; Ping et al., 2020; Xu et al., 2020c). The contribution of CaO_2 hydrolysates to the AD efficiency was $OH^- > •O_2^- > •OH > Ca^{2+}$, with active free radicals ($•O_2^-$ and •OH) exerting an inhibitory effect on the functional enzyme activity of methanogens (Wei et al., 2022). Interestingly, CaO_2 pretreatment accelerates the decomposition of humus and lignocellulose (Wang et al., 2019), as well as the removal of recalcitrant trace organic pollutants (Zheng et al., 2019; Zheng et al., 2021; Zhang et al., 2015b). It is inductively found that the optimal dose typically falls within the range of 0.14–0.20 g/g VS, resulting in about 30–50 % increase in methane yield (Table 2) (Wei et al., 2022; Wang et al., 2019; Wang et al., 2022c). However, if the pH of pretreated sludge is not adjusted from alkaline to neutral after CaO_2 (below 0.20 g/g VS) pretreatment, methanogenesis will be inhibited in AD, compared with unpretreated group (Zheng et al., 2021). Analogous to H_2O_2, the synergistic applications of CaO_2 with other technologies enhance pretreatment performance and AD efficiency, ultimately leading to the reduction of CaO_2 dose. For example, the combined pretreatment of CaO_2 (0.1 g/g VS) and ultrasonic (1 W/mL, 10 min) resulted in 26 % increase in SCOD and 69 % surge in methane yield, compared to the sole CaO_2 pretreatment (Li et al., 2019). Wang and Li (2016) also found that combining CaO_2 (0.1 g/g VS) and microwave (480 W, 2 min) pretreatment resulted in a 2-fold rise in SCOD and an 80.2 % increase in methane yield over the control group. Therefore, considering the high price of CaO_2 and the need for acid conditioning after CaO_2 pretreatment, its combination with other pretreatment technologies to reduce the CaO_2 dose may be more practical.

3.3. Ozone pretreatment

Ozone, with its potent oxidation capacity, can effectively sterilize microorganisms and oxidatively decompose macromolecular organic matters in EPS (Zhen et al., 2017). Furthermore, the hydrolysis of organic components such as protein, polysaccharide, and lipid is enhanced through ozone pretreatment (Du et al., 2021). The principal component analysis revealed that ozone pretreatment contributed to the degradation of aromatic proteins and aromatic amino acids (such as tryptophan and tyrosine), while concurrently causing an elevation in fulvic acid and humic acid with increased pretreatment duration and ozone dose (Du et al., 2021). Studies on the effect of ozone pretreatment on the AD of sewage sludge are summarized in Table 2. Sowmya Packyam et al. (2015) observed the SCOD initially increased and then declined with escalating ozone dosage, with the five-fold increase in SCOD and 165 % enhancement in biogas yield at 1.1 mg O_3/g TS compared to untreated sludge. Ozone dose of less than 100 mg O_3/ g VS was preferred for thickened sludge (20–50 mg TS/L) (Hodaei et al., 2021; Catenacci et al., 2022; Tuncay et al., 2022). The methane yield (128–204 mL/g VS) in AD exhibited an upward trend in response to ozone dose (0–90 mg O_3/g VS), while its growth was limited if further increasing ozone dose (Catenacci et al., 2022). Moreover, several studies have highlighted that the energy and economic evaluation of ozone pretreatment for sewage sludge in AD is unfavorable due to substantial energy consumption and high ozone-related expenses (Khanh Nguyen et al., 2021; Sowmya Packyam et al., 2015; Hodaei et al., 2021; Catenacci et al., 2022).

Table 2
Enhancement of sludge solubilization and AD efficiency by various mono chemical pretreatments.

Pretreatment	Sludge type	Raw sludge property	Pretreatment conditions	Pretreatment performance	Enhancement of AD efficiency	References
Acid	WAS	5.65 % TS, 4.56 % VS; 7.68 g TCOD/L	37 % HCl, pH = 2, 24 h	SCOD increased by about 1.9 times	TCOD removal increased by 7.3 %; VS removal increased by 5.2 %; Biogas yield increased by 12.12 % (231 → 259 mL/g VS); Methane content was about 60 %	(Devlin et al., 2011)
Acid	WAS	2.63 % TS, VS/TS = 54.2 %	4 M HCl, pH = 2.6, 5 h	Soluble TOC increased by 17.9 %	Methane yield increased by 109.3 % (126.8 → 265.4 mL/g VS); Methane content increased by 20 %	(Xu et al., 2022)
Acid	WAS	40.1 g TS/L, 33.2 g VS/L; 54.4 g TCOD/L	FNA, 4.9–6.1 mgN/L, pH = 5, 24 h	DD_{COD} = 8.6 %	VS removal increased by 47.5 %; Methane yield increased by 37.1 % (114.6 → 157.2 mL/g VS)	(Meng et al., 2020)
Acid	WAS	42.6 g TS/L, 33.7 g VS/L; 54.1 g TCOD/L	FNA, 2.13 mgN/L, pH = 5.5, 24 h	DDCOD = 8.5 %	Methane yield increased by 26.9 % (201 → 255 mL/g VS)	(Wang et al., 2013)
Alkali	WAS	8.80 g TS/L, 5.86 g VS/L; 35 mg SCOD/L, 5.83 g TCOD/L	6 M NaOH, pH = 12.50, 24 h	DD_{COD} = 27.8 %		(Xiao et al., 2015)
Alkali	WAS	27.76 g TCOD/L, 185 mg SCOD/L	4 M NaOH, pH = 12.0, 12 h	DD_{COD} = 10.1 %	Methane yield increased by 49.4 % (151.2 → 225.9 mL/g VS); Methane content increased by 11.7 %	(Xu et al., 2020b)
Alkali	WAS	10.6 g TSS/L, 7.7 g VSS/L; 73 mg SCOD/L	4 M NaOH, pH = 10.0, 24 h		TSS removal increased from 13.1 %, VSS removal increased by 12.8 %; Biogas yield increased by 15.4 % (412.5 → 476.0 mL/g VS)	(Shao et al., 2012)
Alkali	WAS	10.07 % TS, VS/TS = 77.30 %; 29.9 g TCOD/L, 196 mg SCOD/L	0.4 % NaOH (*w/v*), 3 h	DD_{COD} = 10.4 %;	Methane yield increased by 48.9 % (266 → 396 mL/g VS)	(Maryam et al., 2021)
Alkali	WAS	22.87 % TS, VS/TS = 47.26 %; 158.3 g TCOD/kg, 1.52 g SCOD/kg	20 mg NaOH/g TS, 24 h	DD_{COD} = 1.83 %	TCOD removal increased by 52.3 %; Methane yield increased by 35 % (74.16 → 100.37 mL/g VS); Methane content ranged from 50 to 60 %	(Zhang et al., 2015a)
Alkali	WAS	19 % TS, VS/TS = 78 %	5 g NaOH/100 g TS, 48 h		Methane yield increased by 39.4 % (109 → 152 mL/g VS); TS and VS removal increased by 45 %, 39.3 %	(Elalami et al., 2020)
Alkali	WAS	20.0 g TS/L, 10.8 g VS/L; 13.66 g TCOD/L, 3052 mg SCOD/L	0.1 mol NaOH/L	DD_{COD} = 26.9 %	Organic degradation increased by 26.4 %; Biogas yield increased by 1.6 % (64 → 65 mL/g VS)	(Li et al., 2012)
Fenton	WAS	30.05 g TS/L, 21.76 g VS/L; 33.84 g TCOD/L, 0.82 g SCOD/L	60 g H_2O_2/Kg TS, 0.07 g Fe_2^+/g H_2O_2, pH = 3, 1 h	TS removal and VS removal increased by 29.2 %, 33.5 %; DD_{COD} = 21.1 %	Methane yield increased by 15.3 % (430 → 496 mL/g VS); Methane content increased by 3.5 %	(Pilli et al., 2016)
Fenton	WAS	10.19 g TS/L, 7.11 g VS/L; 8.50 g TCOD/L, 45 mg SCOD/L	Conventional Fenton pretreatment (CFP:Fe_2^+ + H_2O_2) and Fenton type pretreatment (FTP:Fe + H_2O_2): Catalyst iron dosage = 4 g/Kg TS, H_2O_2 dosage = 40 g/Kg TS, pH = 3, 1 h		TCOD removal in CFP and FTP were 38.4 % and 47.5 %; Methane yield increased by about 38.0 % in CFP and 26.8 % in FTP	(Şahinkaya et al., 2015)
Calcium peroxide	Primary sludge, WAS	Primary sludge: 46.4 g TS/L, 20.2 g VS/L, 43.0 g TCOD/L; WAS: 46.5 g TS/L, 19.6 g VS/L, 45.3 g TCOD/L	0.14 g CaO_2/g VS, 72 h, 25.0 °C	Primary sludge: DD_{COD} = 7.2 %; WAS: DD_{COD} = 6.6 %	Primary sludge: methane yield increased by 6.77 % (405 → 432 mL/g VS); Methane content increased by 2.0 %. WAS: Methane yield increased by 35.1 % (137 → 185 mL/g VS); Methane content increased by 7.0 %	(Wang et al., 2022c)
Calcium peroxide	Primary sludge	27.4 g TSS/L, 20.2 g VSS/L; 30.2 g TCOD/L	0.14 g CaO_2/g VSS, 48 h		VSS removal increased by 25.8 %; Methane yield increased by 34.3 % (306 → 411 mL/g VS)	(Wei et al., 2022)
Calcium peroxide	WAS	19.77 g TSS/L, 11.89 g VSS/L; 14.78 g TCOD/L, 242 mg SCOD/L	0.14 g CaO_2/g VSS, 48 h		Methane yield increased by 47.6 % (146.3 → 215.9 mL/g VS)	(Wang et al., 2019)
Ozone	WAS	12.56 g TS/L, 5.60 g VS/L; 10 g TCOD/L, 400 mg SCOD/L	0.85 mg O_3/min, 1.1 mg O_3/g TS	DD_{COD} = 25.4 %	Biogas yield increased by 164.9 % (202 → 535 mL/g VS)	(Sowmya Packyam et al., 2015)
Ozone	WAS	49.37 g TS/L, 39.89 g VS/L, VS/TS = 80.89 %	62 mg O_3/g VS		TS removal increased by 4.4 %; Biogas yield increased by 7.7 % (219.5 → 236.5 mL/g VS $_{removed}$)	(Hodaei et al., 2021)

4. Biological pretreatment

4.1. Microaeration pretreatment

The low dissolved oxygen environment inherent to microaeration pretreatment augments the metabolic activity of facultative bacteria within sludge, which stimulates the secretion of hydrolases, including amylase, protease, and cellulase (Amin et al., 2021; Merrylin et al., 2013; Nguyen and Khanal, 2018; Zhou et al., 2021). Microaeration pretreatment can be sustained in a continuous reactor system. When the

pretreated sludge enters AD reactors, the facultative bacteria in the sludge can survive and continue to promote sludge hydrolysis. Microaeration not only fosters the enrichment of hydrolytic bacteria but also contributes to enhancing the activity of hydrolases (Ruan et al., 2019). Following a 4-h microaeration pretreatment, Ruan et al. (2019) observed the protease and α-glucosidase activity increased by 64 % and 47 %, respectively. Air flow rate (AFR), temperature (T), and pretreatment time (t) collectively dictate the efficiency of sludge solubilization and AD (Table 3). AFR, as the crucial parameter, typically falls within the moderate range of 0.2–0.4 volume of air per reactor volume per minute (vvm). Montalvo et al. (2016) found that the optimal results, including 36.2 % VS removal and 85.0 % SCOD increase, were achieved at AFR = 0.3 vvm, $T = 35$ °C, and $t = 48$ h, along with a remarkable 111 % increase in methane yield during AD. Similarly, Rashvanlou et al. (2021) reported 65 % elevation in sludge solubilization and 148 % surge in methane yield at AFR = 0.2 vvm and $t = 40$ h, accompanied by increased diversity and activity of the hydrolytic bacteria within AD. Furthermore, the enhancement of methane production in microaeration pretreatment is conducive to achieving energy balance, as its energy demand is comparatively lower than alternative pretreatments, such as ultrasonic and thermal hydrolysis (Zhou et al., 2021; Ruan et al., 2019; Rashvanlou et al., 2021). For example, the energy consumption of microaeration pretreatment of WAS (40 g TS/L) was found to be 59.8 % lower than that of thermal pretreatment (Castillo et al., 2022). As a result, microaeration pretreatment for sewage sludge holds much potential to yield appreciable net energy output in AD.

4.2. Exogenous bacteria pretreatment

Exogenous bacteria are isolated, cultured, and subsequently introduced into sewage sludge under optimized growth conditions, which adeptly release hydrolases to catalyze the decomposition of the organic matters in sludge flocs (Yin et al., 2016). Crucially, neither exogenous methanogenic inhibitors nor negative impacts on the AD systems are introduced by exogenous bacteria pretreatment (Zheng et al., 2014). In an instance involving petroleum refinery sludge, inoculating *Pseudomonas putida* at 35 °C for 10 days, Paul Choudhury et al. (2022) observed a 2.49-fold increase in SCOD, alongside a 55–62 % upswing in methane production during AD. In addition, significant removal effect was observed concerning recalcitrant phenols, oils, and petroleum hydrocarbons. Noteworthily, some thermophilic bacterial strains have been extensively investigated for sludge pretreatment, with their optimal growth temperature typically in the range of 55–65 °C (Table 3) (Kavitha et al., 2016a; Sun et al., 2021; Yang et al., 2015). For instance, when *Bacillus* sp. AT07–1, a thermophilic bacterium, was inoculated at 65 °C for 12 h, Sun et al. (2021) noted substantial reductions of 6.2 times in SCOD, 4.8 times in polysaccharide, and 3.6 times in protein within EPS of WAS. Similar success was achieved with the same thermophilic bacterial strain in the pretreatment of saline waste sludge (Gao et al., 2019a). This suggests that thermophilic bacteria pretreatment effectively disintegrates the sludge structure, followed by the significant release of organic matters. However, it is crucial to recognize the potential impact of thermolysis on sewage sludge while inoculating with thermophilic bacteria.

Surfactants confer the advantage of anti-flocculation. With the assistance of surfactants, the hydrolases secreted by exogenous bacteria could come into more thorough contact with the sludge flocs, resulting in further augmentation of SCOD (Kavitha et al., 2016a). Gao et al. (2019b) observed the combination of thermophilic bacteria and alkyl polyglucose for WAS pretreatment led to an overwhelming increase in SCOD, protein, and polysaccharide, elevating them by 18.7, 25.6, and 12.5 times, respectively. Ushani et al. (2017) and Kavitha et al. (2016a) employed protease-secreting bacteria, assisted by dioctyl sodium succinate and sodium dodecyl sulfate, resulting in the enhancement of 104 % methane and 313 % biogas production in AD, respectively. In conclusion, exogenous bacteria pretreatment of SS significantly enhances AD efficiency.

The introduction of exogenous bacteria presupposes that the activity of indigenous functional microorganisms in the AD system is not impaired. From the batch experiments, the exogenous bacteria carried in the pretreated sludge did not weaken the AD efficiency after entering the AD system, and thus it can be assumed that exogenous bacteria will not inhibit the activity of the anaerobic functional microorganisms. In addition, since the exogenous bacteria normally are facultative or anaerobic, they will have a benign competitive relationship with the indigenous functional microorganisms, and will not reduce the AD efficiency. Moreover, some exogenous bacteria were isolated from the sludge itself, so they could coexist with indigenous bacteria very well when they were reintroduced to the reactors (Yang et al., 2015; Ushani et al., 2017; Kavitha et al., 2014a). Differently, thermophilic bacteria (suitable for temperature around 65 °C) are used in some studies mentioned above, their activity may be reduced in a mesophilic AD system, and thus are vulnerable to competition with indigenous bacteria. However, most biological pretreatments have remained in batch tests, and it is worthwhile to explore whether or how exogenous bacteria can persist as dominant strains in pretreatment and the following AD processes for a long period in continuous reactors. In addition, the expense challenge of identifying and culturing effective bacteria and the potential environmental risks from exogenous bacteria might impede its progress.

4.3. Exogenous hydrolase pretreatment

The efficiency of exogenous hydrolase pretreatment is influenced by the factors such as dose, duration, and enzyme types (Table 3) (Khanh Nguyen et al., 2021; Bahreini et al., 2020; Yu et al., 2013; Pei et al., 2010). Considering the organic composition of sewage sludge, the supplementations of lysozyme, α-amylase, protease, cellulase, and lipase are common practices (Xin et al., 2019; Odnell et al., 2016; Dos Santos Ferreira et al., 2020). Given that protein stands as the primary organic constituent in sewage sludge, the protease pretreatment holds greater performance compared to other hydrolases (Odnell et al., 2016). For example, protease pretreatment exhibited a substantial increase in both SCOD by 75 % and methane yield by 306 % when compared to untreated sludge (Agabo-García et al., 2019). In contrast, the enhancement rates of methane yield were only 73 % and 90 % following cellulase and glucanase pretreatments, respectively. In particular, Wang et al. (2022d) and Lu et al. (2023) discovered that the addition of 500 mg/L trypsin (a crucial hydrolytic enzyme in the AD of WAS) dramatically increased the rate of hydrolysis and acidogenesis, with 140 % higher VFAs production than the control group. Nevertheless, Yu et al. (2013) noted that 129.7 % increase in SCOD and 18.6 % improvement in biogas production with amylase pretreatment, while 75.9 % SCOD and 15.6 % biogas enhancement with protease pretreatment. Besides the reason for the relative high percentage of carbohydrates in the tested sludge, the larger kinetic coefficient of the first-order hydrolysis rate of carbohydrates than proteins may be another cause (Xu et al., 2015; Moreira et al., 2023). Other reasons for the lower effect of protease pretreatment on AD efficiency enhancement could be the accumulation of intermediates of the anaerobic degradation of proteins which inhibited the microorganisms or the attack of proteases to the surface of microorganisms (Müller et al., 2016). Additionally, it is interesting that the performance of mixed hydrolase pretreatment might not surpass that of individual hydrolase (Yu et al., 2013). For instance, Moreira et al. (2023) discovered that the biogas yield of WAS pretreated with a mixture of protease and lipase (205 NmL/g VSS) was lower than that achieved through individual protease (241 NmL/g VSS) and lipase (246 NmL/g VSS) pretreatment. This might be due to the degradation of other hydrolase additives by proteases, reducing their effectiveness in the pretreatment process, because proteases theoretically are able to break down other enzymes that are made up of protein (Tavano, 2013). Hence, the judicious addition of hydrolases holds the opportunity to enhance AD efficiency.

Table 3
Enhancement of sludge solubilization and AD efficiency by various mono biological pretreatments.

Pretreatment	Sludge type	Raw sludge property	Pretreatment conditions	Pretreatment performance	Enhancement of AD efficiency	References
Microaeration	Mixture of primary sludge and WAS	107.1 mg SCOD/L; 3.25 g VSS/L	35 °C, 0.3 vvm, 48 h	VSS decreased by 42.5 %; SCOD increased by 82.6 %	TCOD removal increased by 40.4 %; VSS removal increased by 106.1 %; Methane yield increased by 111 %	(Montalvo et al., 2016)
Microaeration	Mixture of primary sludge and WAS	WAS: 52.03 g TS/L, VS/TS = 83.78 %; 82.95 g TCOD/L, 5.6 g SCOD/L. Primary sludge: 32.13 g TS/L, VS/TS = 76.34 %; 56.00 g TCOD/L, 4.30 g SCOD/L	38 °C, 0.2 vvm, 40 h	DD_{COD} = 4.8 %	Methane yield increased by 147.5 % (120 → 297 mL/g VS)	(Rashvanlou et al., 2021)
Microaeration	Mixture of primary sludge and WAS	41.1 g TS/L, 25.4 g VS/L; 32.8 g TCOD/L, 479 mg SCOD/L	35 °C, 4 vvm, 4 h	DD_{COD} = 0.49 %	VS removal increased by 33.9 %; Biogas yield increased by 16.4 % (93.9 → 109.3 mL/g VS)	(Ruan et al., 2019)
Microaeration	WAS	40.52 g TS/L, 34.57 g VS/L; 113.07 g TCOD/L, 21.35 g SCOD/L	35 °C, 0.35 vvm, 48 h	VS removal increased by 14.74 %; DD_{COD} = 9.0 %	Methane yield increased by 68.4 % (52.9 → 89.1 mL/g VS)	(Castillo et al., 2022)
Microaeration	Mixture of primary sludge and WAS	40.6 g TS/L, 23.7 g VS/L; 29.73 g TCOD/L, 458 mg SCOD/L	35 °C, 4 vvm, 4 h		VS removal increased by 36.1 %; Methane yield increased by 11.7 % (121.2 → 135.4 mL/g VS)	(Zhou et al., 2021)
Exogenous bacteria	Mixture of primary sludge and WAS	WAS: 981.5 mg SCOD/L, 34.4 g TSS/L, 20.5 g VSS/L. Primary sludge: 732.5 mg SCOD/L, 10.8 g TSS/L, 6.1 g VSS/L	Thermophilic bacteria (Bacillus sp. AT07–1, FJ231108), 65 °C, 48 h	SCOD increased by 10.1 times; TSS and VSS removal increased by 27.2 % and 34.3 %		(Sun et al., 2021)
Exogenous bacteria	Petroleum sludge (PS)	11.44 % TS, VS/TS = 86.12 %; 2200 mg SCOD/L	Laccase enzyme-producing bacterium (*Pseudomonas putida* 7525), 35 °C, 10 days	SCOD increased by 2.49 times	Methane yield increased by 54.61 % (20 → 31 L/g VS)	(Paul Choudhury et al., 2022)
Exogenous bacteria	WAS	10.1 g TCOD/L, 150 mg SCOD/L; 7.0 g SS/L, 6.0 g VSS/L	Immobilized protease-secreting bacteria, 35 °C, 36 h	SS removal increased by 17.14 %; DD_{COD} = 4.5 %	Methane production increased by 11.5 times	(Ushani et al., 2017)
Exogenous bacteria	WAS	9.16 g TS/L, 4.22 g VS/L; 10.0 g TCOD/L, 100 mg SCOD/L	thermophilic protease-secreting bacteria (*B. licheniformis*), 55 °C, 24 h	TS removal increased 27 %; DD_{COD} = 24 %	Biogas yield increased by 310.6 % (614 → 2521 mL/g VS)	(Kavitha et al., 2016a)
Exogenous bacteria	WAS	10.5 g TCOD/L, 100 mg SCOD/L; 7.0 g TSS/L, 5.6 g VSS/L	Two strains (Bacillus jerish03, Bacillus jerish04), 40 °C, 42 h	SS removal increased by 25.7 %; DD_{COD} = 19.79 %	Biogas yield increased by 120.3 % (212 → 467 mL/g VS)	(Kavitha et al., 2014b)
Exogenous bacteria	WAS	9.8 g SS/L, 6.0 g VSS/L; 537.5 mg SCOD/L	Thermophilic bacteria (Bacillus sp. AT07–1, FJ231108), 65 °C, 48 h	SCOD increased by 18.72 times		(Gao et al., 2019b)
Exogenous hydrolase	Sewage sludge	4.91 % TS, 2.78 % VS	Glucanase, cellulase, protease; 37 °C, 12 d	SCOD increased by 54.3 %, 67.0 % and 75.0 %	TCOD removal increased by 62.8 %, 111.6 % with addition of cellulase, protease; Methane production increased by 7 %, 7 % and 260 % (20 → 34, 34, 72 mL/g VS)	(Agabo-García et al., 2019)
Exogenous hydrolase	WAS	37.99–38.33 g TS/L, 24.98–27.57 g VS/L; 26.85–30.07 g TCOD/L, 136–211 mg SCOD/L	Amylase, protease and mixture of amylase and protease, 37 °C, 28 h	SCOD increased by 129.7 %, 75.9 % and 82.3 %	Biogas yield increased by 18.6 %, 15.6 %, and 20.2 %	(Yu et al., 2013)
Exogenous hydrolase	WAS	14.86 g TS/L, 11.65 g VS/L; 175 mg SCOD/L	Mixture of lysozyme, protease, α-amylase and cellulase, 37 °C, 3 h	SCOD increased by 29.3 times		(Xin et al., 2019)
Exogenous hydrolase	WAS	5.03 g VSS/L	Protease, lipase and their mixture; 37 °C, 3 h		Biogas yield increased by 41.1 %, 42.3 % and 19.6 % (168 → 237, 239, 201 mL/g VS)	(Moreira et al., 2023)
Exogenous hydrolase	WAS	12.60 g TS/L, 6.74 g VS/L; 28.71 g TCOD/L, 1508 mg SCOD/L	Multi hydrolytic garbage enzyme complex, 30 °C, 24 h	SS reduction was 27.45 %, DD_{COD} = 1.5 %		(Arun and Sivashanmugam, 2018)
Exogenous hydrolase	Sewage sludge	133.37 mg SCOD/L, 8.93 g TCOD/L	Immobilized enzyme mixture (acid protease, pectinase, laccase), 37 °C, 8 h	DD_{COD} = 55.8 %		(Wan et al., 2022)

Exogenous hydrolase pretreatment offers the advantages of convenience and harmlessness, yet it entails a prolonged preparation process, elevated cost, and limited service life (Odnell et al., 2016; Wan et al., 2022). For a continuous pretreatment-AD process, the hydrolases probably need to be dosed all the time to maintain the pretreatment effect, which is not desirable in application.

5. Comparation among different pretreatments

5.1. The effect of pretreatments on sludge solubilization

The dissolution degree of COD (DD_{COD}) is a common metric used to assess the disintegration of the sludge structure and microbial cells through pretreatments, and is calculated as follows:

$$DD_{COD} = (SCOD_P - SCOD_R)/(TCOD_R - SCOD_R) \times 100\% \quad (1)$$

Where $SCOD_P$ is the SCOD in pretreated sludge, $SCOD_R$ and $TCOD_R$ respectively represent SCOD and TCOD in the raw sludge (Fang et al., 2015).

Overall, physical pretreatment exhibited the most pronounced efficiency in the dissolution of organic matters within sludge, succeeded by chemical and biological pretreatments. As shown in Fig. 1a, the mean values of DD_{COD} in physical, chemical, and biological pretreatments stood at 26.49 ± 14.98 %, 14.04 ± 8.91 % and 9.15 ± 8.52 %, respectively. Correspondingly, their respective medians for DD_{COD} were determined to be 27.55 %, 10.10 %, and 4.80 %. This observation underscores that the presence of physical pretreatment increases the availability of substrates suitable for functional microorganisms in AD. This is because physical pretreatment involves substantial energy input and robust mechanical disruption, engendering the consequential fragmentation of EPS and the extensive inactivation and lysis of microbial cells (Tyagi and Lo, 2011). Conversely, the kinetics of organic matter solubilization are relatively sluggish in biological pretreatments, contingent upon the metabolic decomposition of bacteria and catalytic hydrolysis by hydrolases. As the destructive effect induced by biological pretreatment on the microbial cell structure is modest, thereby impeding the release of cytoplasm into the liquid phase (Pei et al., 2010). Therefore, the dissolution efficiency stemming from biological pretreatment markedly lagged that attributed to physical pretreatment. The sludge solubilization effect resulted from chemical pretreatment is intermediate between physical and biological pretreatment, which premised upon chemicals and their hydrolyzed derivatives. Oxidative mechanisms are the most prominent feature of the chemical pretreatment, responsible for the decomposition of sludge organic constituents and inflicted cellular damage (Guan et al., 2018). The chemicals also introduce extreme acidic, alkaline, or thermal environments that engender the structural relaxation of sludge matrix and the microbial deactivation (Xiao et al., 2015).

The DD_{COD} resulting from various mono pretreatments is illustrated in Fig. 1b. Among the physical pretreatments, thermal hydrolysis and high pressure homogenization emerge as the most impactful pretreatments for instigating sludge solubilization, boasting average DD_{COD} values of 39.26 ± 9.30 % and 30.96 ± 5.66 %, respectively. The two pretreatments entail elevated temperature and pressure conditions and furnish heightened energy to sewage sludge, facilitating the effective disruption of both sludge structure and microbial cells (Abelleira-Pereira et al., 2015; Balasundaram et al., 2022; Wahidunnabi and Eskicioglu, 2014). Notably, the solid content of sewage sludge subjected to thermal hydrolysis is greater, which is conducive to direct more concentrated energy flux toward the sludge component and thereby accentuate the dissolution of organic constituents (Barber, 2016). In the terms of chemical pretreatments, Fenton and ozone pretreatments show superior oxidative potential compared to calcium peroxide, consequently yielding higher DD_{COD}. The alkali pretreatment demonstrates a more robust solubilization effect (15.41 ± 10.23 %) than the acid counterpart (8.55 ± 0.05 %). It has been proved that the release of proteins and polysaccharides from sewage sludge in alkaline condition is significantly stronger than that in acidic condition (Chen et al., 2022). Biological pretreatments also exhibited dissimilar performance in sludge solubilization each other, with exogenous bacteria pretreatment proving more effective than microaeration and exogenous hydrolase pretreatments. This discrepancy may be attributed to the continual provisioning of specific high-activity hydrolases by exogenous bacteria. Although endogenous facultative bacteria within microaeration pretreatment can also secrete hydrolases, their growth and metabolism can lead to substrate depletion, potentially impeding the augmentation of SCOD. Additionally, exogenous hydrolases manifest a limited activity period, rendering them insufficient for sustained sludge solubilization (Odnell et al., 2016). In summary, mono pretreatments yield sludge dissolution degrees below 40 % DD_{COD} (Fig. 1b), which highlights the need for further exploration aimed at enhancing the dissolution of organic matters in sewage sludge.

5.2. The enhancement of AD efficiency with pretreatments

The ultimate objective of sludge pretreatment is to enhance AD efficiency, particularly in terms of elevating biogas yield, VS and TCOD removal. Different pretreatments have variable effects on sludge, thereby engendering substantial disparities in AD efficiency.

Biogas yield has emerged as the foremost metric to evaluate AD efficiency. As shown in Fig. 2a, the biological pretreatment exerts the most formidable influence on augmenting biogas yield, surpassing the effects of physical and chemical pretreatments. The average enhancement rates of biogas yield in AD with physical, chemical, and biological pretreatments are 46.12 ± 41.07 %, 40.48 ± 36.84 %, and 71.14 ± 78.89 %, respectively. Concomitantly, their medians were documented as 25.34 %, 35.00 %, and 40.75 %, respectively. Although the solubilization efficiency of biological pretreatment may not rival those of physical and chemical pretreatments, its superior performance in enhancing biogas yield is discernible. This is attributed to the biodegradable nature of the predominantly released organic matter through biological pretreatment, without the substantial release and accumulation of refractory organic substances and toxic compounds when compared to physical and chemical pretreatments (Montalvo et al., 2016; Castillo et al., 2022). In addition, the conditions of the biological pretreatment are relatively mild and less likely to introduce factors that interfere with the AD systems, especially the functional microorganisms. However, it is neglected that biological pretreatment, especially hydrolase pretreatment, adds available COD to the AD system, and it will contribute to the improvement of methane yield. In order to more accurately evaluate the effectiveness of biological pretreatment in increasing methane production from AD of sewage sludge, it is recommended that the theoretical methane production of the exogenous COD be subtracted.

Fig. 2b provides the depiction of the biogas yield ensuing from individual pretreatments. Among physical pretreatments, high pressure homogenization, thermal hydrolysis, ultrasonic, and electric decomposition pretreatments yield enhancement rates of 73.93 ± 44.87 %, 59.68 ± 56.60 %, 50.79 ± 34.49 %, and 43.82 ± 18.88 % in biogas yield, respectively. The observed trends in biogas yield augmentation via physical pretreatments closely parallel those of DD_{COD}. Among chemical pretreatments, ozone pretreatment contributes to a remarkable 90.37 ± 74.57 % increase in biogas yield, while acid, alkali, peroxide, and Fenton pretreatments yield roughly 30 % increments. This outcome can be attributed to the inhibitory effects of chemical residues on the activity of anaerobic microorganisms and functional enzymes (Wei et al., 2022; Wang et al., 2019). Oxygen as by-product of ozone decomposition has more subtle influence on microorganisms and enzymes than other by-products like K^+, Na^+, Ca^{2+}, and Fe^{3+}. Within the domain of biological pretreatments, exogenous bacteria pretreatment, as a potent strategy for enhancing biogas yield, achieves an impressive enhancement rate of 161.84 ± 108.56 %, which is closely associated with its high

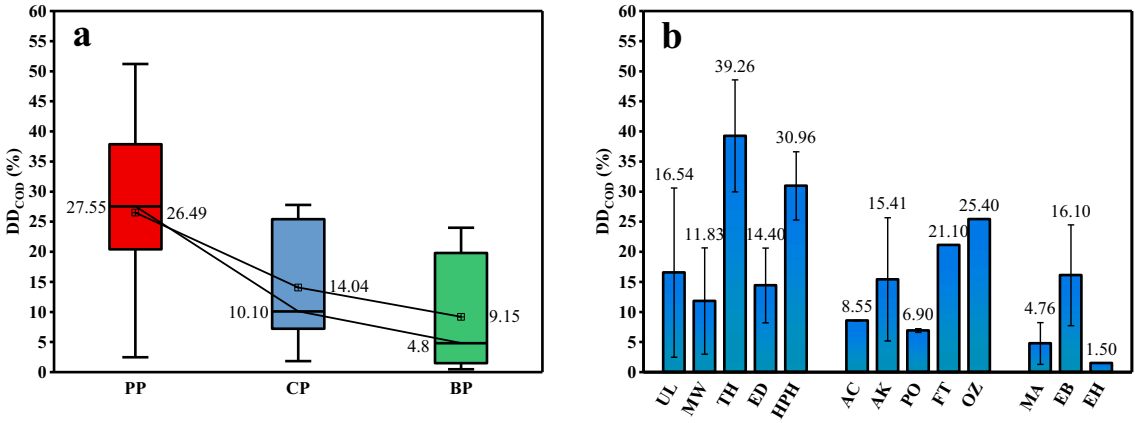


Fig. 1. The dissolution degree of COD (DD_{COD}) in sewage sludge after physical, chemical and biological pretreatments (a) and after various mono pretreatments (b). Notes: The abbreviations of pretreatments correspond to the full names as follows: PP (physical pretreatment), CP (chemical pretreatment), BP (biological pretreatment); UL (ultrasonic), MW (microwave), TH (thermal hydrolysis), ED (electric decomposition), HPH (high pressure homogenization); AC (acid), AK (alkali), PO (calcium peroxide), FT (Fenton), OZ (ozone); MA (microaeration), EB (exogenous bacteria), EH (exogenous hydrolase); The mapping data derives from the literatures summarized in Table 1-Table 3 of this review.

increase in DD_{COD}. Exogenous bacteria pretreatment introduces no inhibitors into the AD systems, thereby ensuring unhindered AD process. Moreover, microaeration pretreatment caused 71.20 ± 52.99 % augmentation in biogas yield, comparable to most physical and chemical pretreatments. In addition, the insignificant effect of exogenous hydrolase pretreatment on enhancing biogas yield is consistent with its low DD_{COD}, which is related to the diminishing enzyme activity and short enzyme lifetime (Odnell et al., 2016).

Organic matters in sewage sludge undergo decomposition and transformation into methane and carbon dioxide during AD, concomitantly leading to the reduction of VS and TCOD. As illustrated in Fig. 2c, physical, chemical, and biological pretreatments yield improvement rates of 52.24 ± 48.02 %, 26.12 ± 15.79 %, and 58.70 ± 33.53 % in VS removal, respectively, coinciding with their performances in the increase of biogas yield. In physical pretreatments, noteworthily, high pressure homogenization, ultrasonic, and thermal hydrolysis pretreatments significantly improve sludge solubilization (Fig. 1b), achieving remarkable enhancements of 138.4 %, 143.3 %, and 131.9 % in VS removal during AD, respectively (Lizama et al., 2018; Zhang et al., 2012a; Han et al., 2017). In contrast, the efficiency of chemical pretreatments in increasing VS removal during AD notably lags behind that of physical and biological pretreatments, aligning with its limited impact on biogas yield enhancement. Among biological pretreatments, microaeration pretreatment emerges as a robust strategy for reducing VS in AD. In the study conducted by Montalvo et al. (2016), microaeration pretreatment made a substantial 106.1 % increase in VS removal during AD. This outcome may be related to the relaxation of sludge structure facilitated by the synergistic effects of microbubbles and facultative bacteria, which magnifies the contact interface between pretreated sludge and anaerobic microorganisms in AD (Zhou et al., 2021).

Correspondingly, the augmentation of TCOD removal is mirrored by the ascending trend in biogas yield and VS removal. The enhancement rates of TCOD removal through physical, chemical, and biological pretreatments are 85.94 ± 82.80 %, 36.38 ± 17.51 %, and 71.60 ± 39.27 %, respectively. The efficiency of biological pretreatment in enhancing TCOD removal slightly trailed that of physical pretreatment. This minor variance is attributed to the concurrent increase of organic matter through biological pretreatment, resulting in the supplementing of TCOD. Specific causes may be that the quantity of anaerobic microorganisms increases due to the absence of inhibitory factors in AD with biological pretreatment, and the addition of exogenous bacteria and hydrolase also further complements the TCOD in AD. In summary, biological pretreatment stands out as an advantageous approach in enhancing AD efficiency of sewage sludge, probably surpassing the benefits offered by physical and chemical pretreatments.

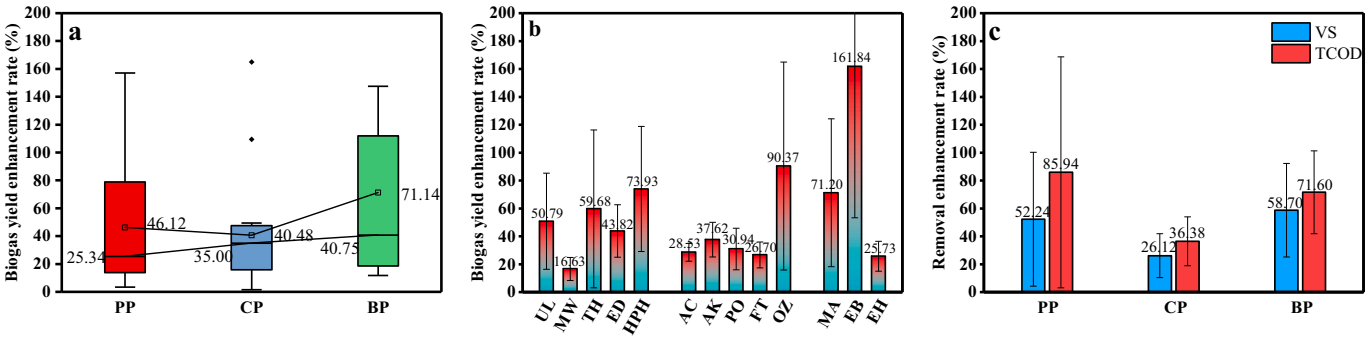


Fig. 2. The enhancement rate of biogas yield in AD of sewage sludge with physical, chemical, and biological pretreatments (a) and with various mono pretreatments (b); and the enhancement rate of VS and TCOD removals in AD of sewage sludge with physical, chemical, and biological pretreatments (c). Notes: The abbreviations of pretreatments correspond to the full names as follows: PP (physical pretreatment), CP (chemical pretreatment), BP (biological pretreatment); UL (ultrasonic), MW (microwave), TH (thermal hydrolysis), ED (electric decomposition), HPH (high pressure homogenization); AC (acid), AK (alkali), PO (calcium peroxide), FT (Fenton), OZ (ozone); MA (microaeration), EB (exogenous bacteria), EH (exogenous hydrolase); The mapping data derives from the literatures summarized in Table 1–Table 3 of this review.

5.3. The effect of pretreatments on microbial community in AD systems

Bacteria and archaea play pivotal roles as functional microorganisms in AD systems. When compared to raw sludge, pretreated sludge not only alters the quantity of microorganisms but also promotes the formation of dominant bacteria and archaea in AD. To delve deeper into the alterations within microbial community through pretreatments, we conducted an examination about the changes in the relative abundance at the bacterial phylum and archaeal genus levels (Fig. 3).

At the bacterial phylum level, the prevalent bacteria in AD encompasses *Bacteroidetes*, *Firmicutes*, *Proteobacteria*, *Chloroflexi*, *Actinobacteria*, *Synergistetes* et al. (Xu et al., 2021b). Among these bacteria, *Bacteroidetes*, *Firmicutes*, *Proteobacteria*, and *Chloroflexi* stand out as the most frequently dominant phyla in AD (Fig. 3a). *Bacteroidetes* and *Firmicutes* are recognized for secreting extracellular hydrolases, instrumental in the decomposition of macromolecular organic compounds. Noteworthy increase in the relative abundance of *Bacteroidetes* (by 33.3 %) and *Firmicutes* (by 55.9 %) has been observed in AD of sewage sludge subjected to alkaline pretreatment (Xu et al., 2020b). *Chloroflexi* and *Proteobacteria*, crucial players in protein and polysaccharide hydrolysis and acidification, accounted for 34 % and 13 % of the relative abundance in AD of sewage sludge pretreated by high pressure homogenization, respectively (Nabi et al., 2021). In the study of Chen et al. (2022), the relative abundances of *Bacteroidetes*, *Firmicutes*, and *Chloroflexi* in AD with acid, alkaline and thermal hydrolysis pretreatments were all significantly enhanced compared to the untreated group, yet with the large decrease of *Proteobacteria*. Unexpectedly, Fenton pretreatment has a significant attenuating effect on the relative abundance of *Firmicutes*, but an enhancing effect on the abundance of *Proteobacteria*, *Bacteroidetes* and *Chloroflexi* (Hou et al., 2020). It can therefore be surmised that *Firmicutes* and *Proteobacteria* are in intense competition in the AD systems. Overall, however, the physical and chemical pretreatments have a significant enhancing effect on the abundance of *Firmicutes*, but an attenuating effect on the relative abundance of *Proteobacteria* (Fig. 3a). This may be because, *Firmicutes*, which is richer in genus levels of bacteria than *Proteobacteria*, is better adapted to the harsh environment of the AD fed with sludge pretreated by physical and chemical pretreatments (Wu et al., 2022; Li et al., 2020).

Of note, microaeration pretreatment exhibits a marked advantage in the formation of multiple dominant strains in AD systems (Fig. 3a). The research conducted by Ruan et al. (2019) demonstrated higher relative abundance of nearly all functional bacteria in AD with microaeration pretreatment, indicating its positive impact on functional bacteria thrive without inhibitors and harsh environments. In addition, microaeration pretreatment did not significantly alter the structure of the bacteria community, and almost all bacteria phyla were relatively stable in AD (Zhou et al., 2021). This pretreatment contributes to providing VFAs from multiple pathways for methanogens in AD, thereby ensuring enhanced methanogenesis efficiency. However, on the contrary, the exogenous hydrolase pretreatment may tend to cause the increase of particular bacteria at the expense of others. For example, in the study of Wan et al. (2022), the pretreatment with various exogenous hydrolases significantly increased the relative abundance of *Firmicutes* (from 24.50 % to 52.08 %), while concurrently reducing those of *Proteobacteria*, *Chloroflexi*, *Bacteroidetes*. Although the simplification of dominant bacteria might enhance hydrolysis and acidification efficiency, it could also potentially compromise AD systems stability, which is not conducive to methanogenesis. In conclusion, the competition among bacteria, especially *Firmicutes* and *Proteobacteria*, is exacerbated in AD systems fed with pretreated sludge compared to those fed with untreated sludge.

Wang et al. (2022a) provided a detailed classification of methanogens based on taxonomy. Acetotrophic and hydrogenotrophic methanogens emerge as the primary contributors to methane production in AD. At genus level, *Methanosaeta* and *methanothrix* represent obligatory acetotrophic methanogens; *Methanobacterium*, *Methanoculleus*, *Methanospirillum*, *Methanobrevibacter*, *Methanolinea* and *Methanocorpusculum* et al. are indicative of hydrogenotrophic methanogens; *Methanosarcina* is a universal methanogen capable of utilizing acetate, hydrogen and carbon dioxide, and methyl group to produce methane (Xu et al., 2021b). Of these, *Methanosaeta* and *Methanobacterium* are the two commonly dominant methanogens. It is found that most pretreatments foster an increased relative abundance of *Methanosaeta* (Fig. 3b). In comparison to AD fed with raw sludge, the relative abundance of *Methanosaeta* increased by 150.74 %, 34.04 %, 25.00 %, and 38.00 % in AD with ultrasonic, high pressure homogenization, alkali and microaeration pretreatments, respectively (Li et al., 2018; Nabi et al., 2021; Xu et al., 2020b; Zhou et al., 2021). The proliferation of *Methanosaeta* may be attributed to its efficient adhesive and granulation characteristics, resulting in higher resistance to inhibitors in AD systems fed with pretreated sludge than other methanogens (Di Maria, 2017). Given that the liquid phase of pretreated sludge also has significantly higher VFAs levels compared to raw sludge, the conditions are conducive to the dominance of *Methanosaeta*. Although hydrogenotrophic methanogens comprise a greater diversity of species, their total relative

a

Bacteria	UL[1]		UL[2]		MW[3]		TH[4]		TH[5]		HPH[6]		AC[7]		AK[8]		AK[9]		FT[10]		MA[11]		MA[12]	
	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD
Firmicutes			22.2	15.7			36.5	68.2	3.0	7.4	1.1	1.9	11.7	16.9	19.2	28.8	17.1	23.9	70.8	7.0	4.4	5.6	3.4	4.0
Chloroflexi			21.1	21.9			1.9	0.9	14.4	9.8	37.0	34.0	17.1	16.5	7.1	4.4	13.2	27.6	0.0	10.2	12.7	17.5	13.4	21.2
Proteobacteria			15.8	10.2			8.4	9.0	34.1	20.8	16.0	13.0	18.1	18.3	15.4	10.6	17.0	13.0	10.9	33.1	22.0	25.4	34.9	29.4
Bacteroidetes							11.2	9.6	30.3	26.0	7.0	9.0	7.9	8.0	20.7	27.9			15.3	33.9	10.4	10.5	12.2	6.6
Actinobacteria			13.4	9.3			6.0	3.0			6.0	5.0	9.8	11.7	8.6	7.7	12.8	19.9	2.5	8.7	4.3	6.8	7.4	6.2
Synergistetes							1.9	1.9			3.8	2.2	11.6	9.1	2.8	1.1					4.2	7.4	2.3	2.5

b

Archaea	UL[1]		UL[2]		MW[3]		TH[4]		TH[5]		HPH[6]		AC[7]		AK[8]		AK[9]		FT[10]		MA[11]		MA[12]	
	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD	Raw-AD	Pre-AD
Methanosaeta[a]	27.0	67.7			7.0	8.1			15.7	27.4	47.0	63.0	42.5	38.7	41.2	51.5	42.6	2.0	0.0	37.3	66.4	68.4	49.0	67.6
Methanosarcina[b]					9.6	21.2							10.9	9.3	0.8	3.1	6.4	63.3	0.0	9.7				
Methanobacterium[c]					0.3	0.4			2.6	5.1	39.0	19.0	23.1	30.0	47.7	37.2	8.5	16.3	5.0	12.3	8.0	8.8	44.4	26.4
Methanoculleus[c]					50.4	47.3							2.4	2.0										
Methanospirillum[c]															2.2	2.0	27.7	4.1	0.0	3.9				
Methanocorpusculum[c]	55.9	6.7																						
Methanobrevibacter[c]													2.4	2.7	2.7	1.5			89.1	22.8	2.6	4.3		

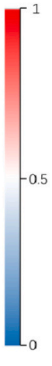


Fig. 3. The relative abundance of the main functional microorganisms in AD systems of raw sludge (Raw-AD) and pretreated sludge (Pre-AD): the phylum level of bacteria (a), the genus level of archaea (b). Notes: The color shade indicates the position of the cell value in relation to all the values in the graph; Blanks and "0.0" indicate that the microorganism not detected, or not available, or with negligibly low abundance; The abbreviations of pretreatments correspond to the full names as follows: UL (ultrasonic), MW (microwave), TH (thermal hydrolysis), HPH (high pressure homogenization); AC (acid), AK (alkali), FT (Fenton), MA (microaeration); The symbol superscripts of "a", "b" and "c" in (b) represent acetotrophic, polytrophic and hydrogenotrophic methanogens, respectively; The superscripts correspond to the following literatures:1- (Li et al., 2018); 2- (Zheng et al., 2023); 3- (Zhang et al., 2016); 4- (Chen et al., 2018); 5- (Niu et al., 2020); 6- (Nabi et al., 2021); 7-(Xu et al., 2021a); 8- (Xu et al., 2020b); 9- (Sun et al., 2014); 10- (Hou et al., 2020); 11- (Ruan et al., 2019); 12-(Zhou et al., 2021).

Table 4
Energy and economy data in the AD of sewage sludge with various pretreatments.

Pretreatment	Sludge type	Pretreatment conditions	Energy and economic analysis	References
Ultrasonic	WAS	24 kHz, 0.73 W/mL, 10 min	Net energy output decreased from 533 to −4063 J/g TS	(Çelebi et al., 2021)
Microwave	Mixture of fermented primary sludge and WAS	2450 MHz, 30 min, 160 °C, 28.12 KJ/g TS	Net energy output decreased from 5.5 to −28.4 KJ/g VS	(Kor-Bicakci et al., 2019)
Thermal hydrolysis	Digested sludge	180 °C, 30 min	Net energy output was 372 KW·h/t DS in AD-THP	(Cai et al., 2021)
Thermal hydrolysis	WAS	200 °C, 30 min	Net energy output decreased from 25 to −2704 KJ/g WAS	(Yang et al., 2010)
Electric decomposition	Petroleum sludge	60 V, 83.5 min, 11.6 cm	Net energy output increased by 10.9 times	(Paul Choudhury and Kalamdhad, 2021)
Electric decomposition	WAS	15 V, 1 h	The cost saved 8.10 $/m^3 WAS	(Zeng et al., 2019)
Electric decomposition	WAS	20 V, 5 cm, 30 min	Specific energy input was 1.811 kJ/g TS	(Ye et al., 2016)
High pressure homogenization	WAS	12,000 psi	Net energy output decreased from 4.76 to 4.48 KJ/g COD	(Wahidunnabi and Eskicioglu, 2014)
Acid	WAS	37 % HCl, pH = 2, 24 h	The cost increased from 0.58 to 5.28 £/t WAS	(Devlin et al., 2011)
Acid	WAS	4 M HCl, pH = 2.6, 5 h	Economic benefit increased by 162 %; the carbon emissions reduced by 2.63 × 10^7 kg CO_2/year	(Xu et al., 2022)
Fenton	WAS	60 g H_2O_2/kg TS, 0.07 g Fe^{2+}/g H_2O_2, pH = 3, 1 h	Net energy output increased from 91 to 285 kWh/t TS	(Pilli et al., 2016)
Ozone	WAS	0.85 mg O_3 /min	Net energy output was −25.7 kWh/t WAS; Net cost was 5.27 USD/t WAS	(Sowmya Packyam et al., 2015)
Ozone	WAS	0.05 and 0.1 g O_3 g/TS	Net energy output decreased from 272.7 to −56.1 kJ/g TS	(Hodaei et al., 2021)
Microaeration	Mixture of primary sludge and WAS	38 °C, 0.2 vvm, 40 h	Net energy increased from 0.56 to 1.2 kWh/kg VS	(Rashvanlou et al., 2021)
Microaeration	Mixture of primary sludge and WAS	35 °C, 4 vvm, 4 h	The cost saved 7.5 €/t VS	(Ruan et al., 2019)
Microaeration	WAS	35 °C, 0.35 vvm, 48 h	60 % less energy than thermal pretreatment	(Castillo et al., 2022)
Microaeration	Mixture of primary sludge and WAS	35 °C, 4 vvm, 4 h	The cost saved 7.9 €/t VS	(Zhou et al., 2021)

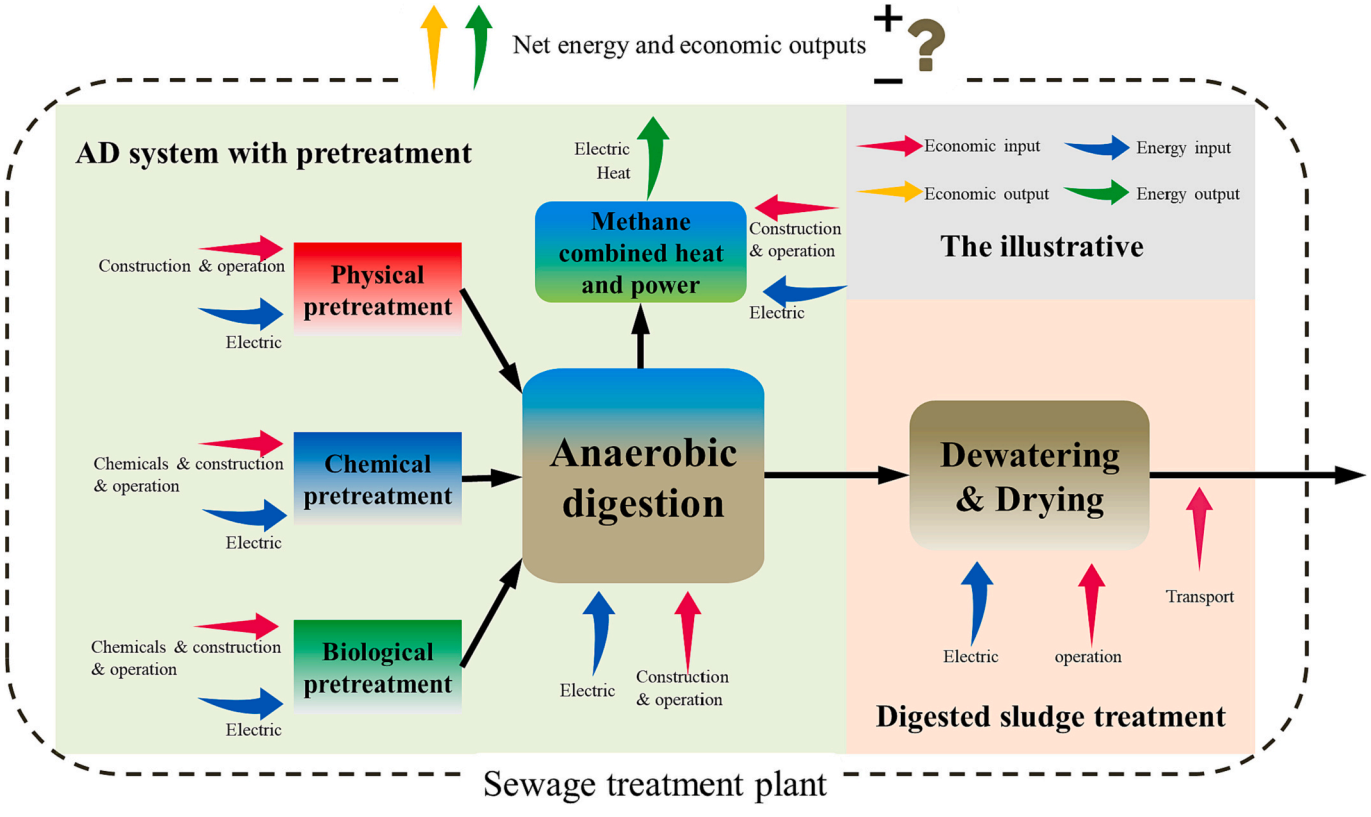


Fig. 4. A full energy and economy assessment route suggested for pretreatment-AD systems in sewage treatment plants.

abundance is not as substantial as that of acetotrophic methanogens. Even the relative abundance of hydrogenotrophic methanogens in AD systems with pretreatment tends to decline (Fig. 3b). Specifically, compared to AD systems without pretreatment, the relative abundance of *Methanobacterium* diminished by 51.28 %, 22.01 %, and 40.54 % following high pressure homogenization, alkali, and microaeration pretreatment, respectively (Nabi et al., 2021; Xu et al., 2020b; Zhou et al., 2021). This trend indicates that hydrogenotrophic methanogens

struggle to compete with acetotrophic methanogens in AD systems fed with pretreated sludge. Differently, Xu et al. (2021a) found the reducing interfacial free energy and enhancing electron transfer capacity of sewage sludge through isoelectric point pretreatment, reinforced the biogas attachment and reaction on the sludge surface, which favored the 29.9 % abundance increase in *Methanobacterium*. This may also be the reason that some studies on sludge pretreatment have indicated a minor rise in the abundance of *Methanobacterium*, while it is still overall lower than that of *Methanosaeta* (Ruan et al., 2019; Hou et al., 2020; Wang et al., 2022e; Niu et al., 2020).

5.4. Energy and economic analysis of AD with pretreatments

The energy and economic performances of AD fed with pretreated sludge are summarized in Table 4. It is notable that most pretreatments failed to effectively enhance the net energy output of AD and even resulted in a negative net energy output. Physical pretreatment relies on mechanical equipment to provide the energy required for promoting the sludge disintegration, which often leads to unexpected energy loss during the pretreatment stage. For instance, ultrasonic, microwave and thermal hydrolysis pretreatment exhibited negative net energy output, measuring at −4063 J/g TS, −8.4 kJ/g VS and − 2704 kJ/kg WAS, respectively, despite their significant improvement in sludge solubilization and AD efficiency (Kor-Bicakci et al., 2019; Yang et al., 2010; Çelebi et al., 2021). In contrast, it is worth noting that electric decomposition pretreatment demonstrates the potential to yield a positive net energy output. Ye et al. (2016) demonstrated that the specific energy input for electric decomposition pretreatment can be as low as 1.811 kJ/g TS, significantly less than that required for microwave, ultrasonic, and thermal hydrolysis pretreatments. And the research conducted by Paul Choudhury and Kalamdhad (2021) exhibited a 10.9-fold increase in net energy output, likely attributable to the low energy demand of electric decomposition pretreatment itself. In this method, electric energy is directly supplied to sewage sludge without the conversion to mechanical energy, thereby minimizing energy loss. Moreover, as mentioned above, the AD systems with electric decomposition pretreatment demonstrated notable enhancement in biogas yield by 43.82 ± 18.88 %. Due to the high energy consumption in chemical production processes, most chemical pretreatments are also proven challenging to achieve a positive energy output (Sowmya Packyam et al., 2015; Hodaei et al., 2021). Moreover, chemical pretreatment introduces additional environmental risks that may increase the difficulty and expense of the subsequent disposal of digested sludge. Likewise, among biological pretreatments, the strategies of dosing exogenous bacteria and hydrolases have similar impediments. In contrast, microaeration pretreatment has exhibited the potential for yielding a positive energy balance, thanks to their inherent benefits including low energy consumption and a significant increase in biogas yield. Building on these merits, Rashvanlou et al. (2021) demonstrated an impressive 114.3 % increase in net energy output through the implementation of microaeration pretreatment. Notably, when compared to thermal hydrolysis pretreatment, microaeration pretreatment achieved remarkable energy savings, cutting energy consumption by as much as 60 % (Castillo et al., 2022).

There is no doubt that the economic benefits derived from AD are positively linked to the net energy output. Based on the energy analysis above, AD with electric and microaerobic pretreatments exhibited the potential to yield cost savings. In a study by Zeng et al. (2019), electric decomposition pretreatment translated to a savings of $8.10/m^3 WAS. Similarly, microaeration pretreatment also led to cost savings of 7.5–7.9 €/t VS (Zhou et al., 2021; Ruan et al., 2019). For most physical pretreatments, particularly thermal hydrolysis and ultrasonic, however, their negative net energy outputs normally translated to unfavorable economic returns, and even economic loss in some cases (Mainardis et al., 2021; Kim et al., 2022; Dhar et al., 2012; Kavitha et al., 2016b; Peng et al., 2018). In addition, for those AD processes incorporating pretreatments involving the addition of chemicals, exogenous bacteria, and hydrolases, their economic benefits may be largely negative due to the high production costs involved (Devlin et al., 2011; Sowmya Packyam et al., 2015). Consequently, those pretreatments that yield negative economic outcomes face challenges in practical application. In summary, sludge pretreatment is a high-energy and high-cost process, yet with electric and microaerobic pretreatments holding the potential to yield positive energy and economic outcomes as well as being green and harmless.

However, there are relatively few studies on energy and economy assessment, moreover, the scope of the assessment is not yet uniform and comprehensive. This leads to discrepancies and inaccuracies in the energy and economic benefits of AD with sludge pretreatment. Whether or how much AD with pretreatment can improve the net energy and economic output in sludge treatment and disposal still needs to be explored. Uthirakrishnan et al. (2022) have meticulously outlined specific calculation methodologies and provided comprehensive case studies for assessing energy input and output, and the economic feasibility of the pretreatment-AD processes, which offers a valuable reference for the evaluation of the pretreatment-AD system. In order to obtain a more accurate and comparable assessment, a full energy and economic assessment for AD systems assisted by pretreatment in sewage treatment plants is proposed (Fig. 4). Firstly, the energy consumption and economic input of pretreatments as well as AD itself are specified. Then, the methane produced by AD of sewage sludge is recovered in the form of combined heat and power, which should be subtracted. In addition, due to the changes in the properties of the digested sludge like solid content, viscosity and calorific value, the impact on subsequent dewatering, drying and final disposal is significantly different with raw sludge (Çelebi et al., 2021). Thus, the energy and economic assessment for the treatment of digested sludge is worth to be executed.

6. Challenges and perspectives

At present, sludge pretreatments and AD still confront many challenges, and future research should be focused.

(1) Current pretreatment studies are primarily lab-scale batch tests, lacking long-term monitoring and practical methane utilization, which limits the assessment of energy and economic viability. Developing semi-continuous and continuous systems with methane capture could enhance the technology's application.
(2) The lack of standardized pretreatment conditions hinders in-depth research and practical application. Standardizing operational parameters such as sludge solids content, energy input, and exogenous substance dose is crucial for effective pretreatments.
(3) While microbiology has contributed to understanding AD mechanisms, its role in monitoring AD dynamics is underutilized. Establishing microbial communities as monitoring indicators could facilitate AD system adjustments during fluctuations.
(4) Electric decomposition and microaeration pretreatments show promise but face practical challenges, such as scaling up equipment safety and efficiency, and preventing air diffuser clogging. Optimizing energy-efficient pretreatments and addressing these issues are essential. In addition, combining microaeration with electric decomposition may be a promising direction.

7. Conclusions

Different pretreatments have various effects on sludge solubilization and AD efficiency. The sludge solubilization performance is in the order of physical, chemical, and biological pretreatments, while with each below 40 % DD_{COD}. However, in terms of enhancing AD efficiency, biological pretreatment outperforms physical and chemical pretreatments, especially microaeration pretreatment and exogenous bacteria pretreatment. Pretreatments boost the competition of functional microorganisms in AD systems, leading to significant changes in

microbial community. Emphatically, pretreatments normally elevate the relative abundance of *Firmicutes* and *Methanosaeta* (acetotrophic methanogens) while decreasing that of *Proteobacteria* and *Methanobacterium* (hydrogenotrophic methanogens). Despite enhancing AD efficiency, however, most pretreatments yield negative energy and economic outcomes, except for electric decomposition and microaeration pretreatment. Future research should evaluate the financial viability and environmental sustainability of large-scale sludge pretreatment-AD systems using techno-economic analysis or life cycle assessment. Addressing the practical challenges of promising pretreatments like electric decomposition and microaeration is also vital. Microbial communities can serve as valuable indicators for rapid regulation of AD system.

CRediT authorship contribution statement

Pan Zhou: Writing – review & editing, Writing – original draft, Visualization, Data curation, Conceptualization. **Dunjie Li:** Writing – review & editing, Conceptualization. **Cong Zhang:** Writing – review & editing. **Qian Ping:** Writing – review & editing. **Lin Wang:** Writing – review & editing. **Yongmei Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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